

In the name of Allah

Review Question.

Q.1: Define the following terms: data, database, DBMS, database system, data base catalog, program-data independence, user view, DBA, end user, canned transaction, deductive database system, persistent object, meta-data, and transaction-processing application.

A.1: Data is called raw materials. Like numbers 10, 20, 30 this is data.

Database: The simple definition of Database is collection of related data.

10, 20, 30 if we add or give average from these three number it is database.

DBMS: Is the database management system is one of software of database.

Which we can store our data on it.

Database System: Define a database involves specifying the database, structure, and constraints of the data to be stored in the database. A database may be generated and maintained manually or it may be computerized.

Database Catalog: A catalog is going to take a look at all the different types of data you have your organization and put it together and organize it into one place.

Program- data independence: In traditional file processing, the structure of data files is embedded in the application programs, so any changes to the structure of a file may require changing all programs that access that file. By contrast, DBMS access programs do not require such changes in most cases. The structure of data files is stored in the DBMS catalog separately from the access programs. We call this property program-data independence.

User view: A database typically has many types of users, each of whom may require a different perspective or view of the database. A view may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored. A multiuser DBMS whose users have a variety of distinct applications must provide facilities for defining multiple views.

DBA: The DBA is responsible for authorizing access to the database, coordinating and monitoring its use, and acquiring software and hardware resources as needed. The DBA is accountable for problems such as security breaches and poor

system response time. In large organizations, the DBA is assisted by a staff that carries out these functions.

End user: End users are the people whose jobs require access to the database for querying, updating, and generating reports; the database primarily exists for their use. There are several categories of end users: Casual end users, Naive or parametric end users, Sophisticated end users, Standalone users.

Canned Transaction: Using standard types of queries and updates called canned transaction.

Deductive database system: Some database systems provide capabilities for defining deduction rules for inferencing new information from the stored database facts. Such systems are called deductive database systems.

Persistent object: Databases can be used to provide persistent storage for program objects and data structures. This is one of the main reasons for object oriented database systems.

Meta-data: The database definition or descriptive information is also stored by the DBMS in the form of a database catalog or dictionary; it is called meta-data.

Transaction-processing application: An application program accesses the database by sending queries or requests for data to the DBMS. A query typically causes some data to be retrieved; a transaction may cause some data to be read and some data to be written into the database.

Q.2: What four main types of actions involve databases? Briefly discuss each.

A.2: Requirements specification and analysis, Conceptual design, Logical design and Physical design.

Design of a new application for an existing database or design of a brand new database starts off with a phase called Requirements specification and analysis. Conceptual design that can be represented and manipulated using some computerized tools so that it can be easily maintained, modified, and transformed into a database implementation.

The design is then translated to a logical design that can be expressed in a data model implemented in a commercial DBMS.

The final stage is physical design, during which further specifications are provided for storing and accessing the database.

Q.3: Discuss the main characteristics of the database approach and how it differs from traditional file systems.

A.4: In traditional file processing, each user defines and implements the files needed for a specific software application as part of programming the application. And the main characteristics of the database are following:

Self-describing nature of a database system.

Insulation between programs and data, and data abstraction.

Support of multiple views of the data.

Sharing of data and multiuser transaction processing.

Q.4: What are the responsibilities of the DBA and the database designers?

A.4: The DBA is Database Administrator. The DBA is responsible for authorizing access to the database, coordinating and monitoring its use, and acquiring software and hardware resources as needed. The DBA is accountable for problems such as security breaches and poor system response time. In large organizations, the DBA is assisted by a staff that carries out these functions. Database designers are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data.

Q.5: What are the different types of database end users? Discuss the main activities of each.

A.5: 1: Casual end users: They use a sophisticated database query interface to specify their requests and are typically middle- or high-level managers or other occasional browsers.

2: Naive or parametric end users: Their main job function revolves around constantly querying and updating the database, using standard types of queries and updates called canned transactions.

3: Sophisticated end users: include engineers, scientists, business analysts, and others who thoroughly familiarize themselves with the facilities of the DBMS in order to implement their own applications to meet their complex requirements.

4: Standalone users: maintain personal databases by using ready-made program

packages that provide easy-to-use menu-based or graphics-based interfaces. An example is the user of a financial software package that stores a variety of personal financial data.

Q.6: Discuss the capabilities that should be provided by a DBMS.

A.6: The DBMS is a general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications.

A.7: Discuss the differences between database systems and information retrieval systems.

A.7: The Database systems propose is: to store, manage and retrieve structured data. Like the bank system, student registration system and etc.

And the information retrieval system purpose is to search and find relevant information from unstructured data.

Exercises

Q.8: Identify some informal queries and update operations that you would expect to apply to the database shown in Figure 1.2.

A.8:

Q.9: What is the difference between controlled and uncontrolled redundancy? Illustrate with examples.

A.9:

Q.10: Specify all the relationships among the records of the database shown in Figure 1.2.

A.10: The relationship between them are major, department, credit hours, and etc. and that element are unique are: Name, Student number, Instructor.

Q.11: Give some additional views that may be needed by other user groups for the database shown in Figure 1.2.

A.11:

Q.12: Cite some examples of integrity constraints that you think can apply to the database shown in Figure 1.2.

A.12:

Q.13: Give examples of systems in which it may make sense to use traditional file processing instead of a database approach.

A.13: Like in past in government store the data in paper and they made a lot of problem like the name of people their properties and that made redundancy.

Q.14: Consider Figure 1.2

a. If the name of the 'CS' (Computer Science) Department changes to 'CSSE' (Computer Science and Software Engineering) Department and the corresponding prefix for the course number also changes, identify the columns in the database that would need to be updated.

b. Can you restructure the columns in the COURSE, SECTION, and PREREQUISITE tables so that only one column will need to be updated?

A.14: a: Three column should be change first the column of Major, the second column should be change is Department, and the third one column is course number.

b: So in all of files the course number are same so we update the course number.

Ahmadzai